

Forecasting Electric Demand of Distribution System Planning in Rural and Sparsely Populated Regions

H. Lee Willis, Fellow
ABB Automated Distribution Division
1021 Main Campus Drive
Raleigh, NC 27606

Linda A. Finley, Senior Member
Snohomish County PUD
P. O. Box 1107
Everett, WA 98206

Michael J. Buri, Member
ABB Automated Distribution Division
1021 Main Campus Drive
Raleigh, NC 27606

ABSTRACT

Modern computerized distribution load forecasting methods, although accurate when applied to urban areas, give somewhat less satisfactory results when forecasting load growth in sparsely populated rural areas. This paper examines the differences between rural and urban load growth histories, identifying a major difference in the observed behavior of load growth. This difference is exploited in a new simulation forecasting algorithm. Tests show the new method is as accurate in forecasting rural load growth and as useful for analyzing DSM impacts than past methods, while requiring considerably lower computer resources and data than other simulation methods of comparable accuracy.

INTRODUCTION

One of the first steps in planning the expansion of a distribution system is the forecast of the future loads which the system is expected to serve. This distribution load forecast defines many of the requirements for the future system, including the overall capacities needed for substations and feeders throughout the system. Therefore, it is quite important that this forecast be done accurately and in accordance with the planning needs. In addition, the comprehensive integrated resource T&D plans required of many utilities to assess possible savings from DSM, need a similar forecast of load by location, a requirement met by the load analysis and forecasting.

The last decade has seen considerable progress in the development of computerized distribution load forecasting methods [1-11] and an extension of these methods to T&D Demand-Side Management (DSM) assessment [11]. Most of these methods are designed for application to densely populated regions, with their computerized techniques, data needs, and planning goals tailored to urban situations [1-11]. They have not proven completely suitable for application to sparsely populated regions. This paper describes a new forecasting method that is particularly appropriate for application to rural or sparsely populated regions. It also examines rural load growth and highlights several basic differences in load growth between rural and urban areas.

This paper will appeal predominantly to engineers and planners working in the planning and design of rural electric distribution systems, as well as those interested in the theory and methodology of planning. It begins with a brief review of distribution load forecasting methods. The authors then discuss the specific differences between electric load density and growth behavior in urban and rural regions, as observed in a series of tests. This examination of load growth data from several utilities reveals that there is a considerable difference in how load growth appears in rural versus urban regions. When applied to rural distribution load forecasting, existing simulation methods require excessive computer time and give relatively poor accuracy. The structure and data requirements of a new forecasting method tailored to rural applications is explained. Tests on three utility systems show the new method outperforms previous forecast algorithms in accuracy, data requirements, and numerical computation time.

FORECASTING AT THE DISTRIBUTION LEVEL

The load forecast used in distribution planning must project not only the future loads, but also their locations, in sufficient geographic detail to allow the planner to locate and size distribution equipment additions such as

substations, feeders, and feeder segments [1, 11]. Ideally, the planner needs a map of future loads, as illustrated in Figure 1. This section will only summarize existing methodology for developing this type of forecast. A thorough discussion is given in reference 11.

In all prior forecasting methods, the "where" aspect of distribution load forecasting is handled by dividing the region into a number of small areas and forecasting the future load for each. The most accurate and widely used computerized techniques use a grid of uniformly sized squares to define these small areas [1, 4-11]. Typically, small area size is somewhere in the neighborhood of 40 acres (squares 1/4 mile to a side). Fig. 1 is composed of 39,000 square areas, each 1/5 mile wide. Computer programs to forecast up to 525,000 such areas in a single forecast have been developed [4].

Distribution forecasting methods for small areas fall into two categories. *Trending methods* forecast future load growth based only on extrapolation of past load growth trends. For example, the annual peak loads for the past 10 years for each feeder in the system may be used to develop a forecast of its future peak loads using a polynomial curve fit [1, 2]. There are a wide range of other methods, some applying non-algebraic methods [6]. But regardless of the algorithm or approach, trending methods invariably have three characteristics. First, they are relatively economical to apply, usually requiring only feeder load metering and customer billing data, all easily available. Second, computational needs are very low—many methods can be implemented by hand calculators. Third, trending, no matter what the specific method, has proven inaccurate beyond the very short range, two to four year period [8].

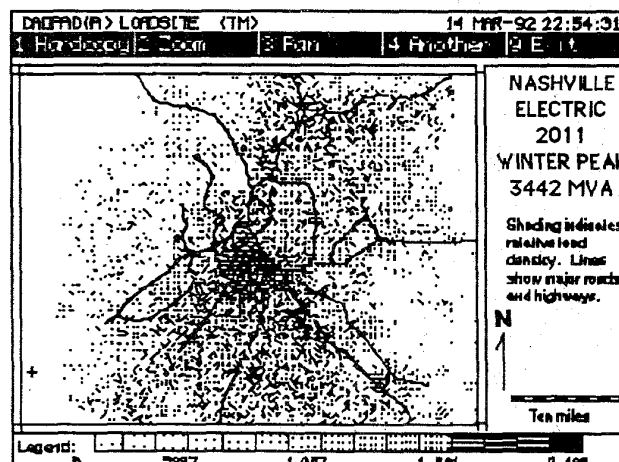


Figure 1. A distribution load forecast produces a map of electric load density, as shown here in a year 2011 projection for a medium sized city. A set of forecast maps like this shows where and how much load growth is expected, and information to help plan location and capacity of T&D facilities.

Land Use Simulation

By contrast, simulation techniques of the type discussed in this paper have proven considerably more accurate, particularly in the five-year or more time frame, where there error is often only 1/7 of that given by the very best trending methods [8]. However, this entails a consequent increase in data and computer resources [1, 8, 11]. Simulation methods base their forecast on land-use, or customer class, distinctions by location within the utility

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service territory. While algorithms differ considerably, all basically attempt to predict where future residential, commercial and industrial load will grow, on a small area basis. The authors believe land-use simulation is the only realistic choice for long-range distribution planning studies, where accuracy and multi-scenario planning capability are necessary [1, 4, 11]. In addition, simulation methods have proven to be the only useful manner of forecasting DSM impacts at the T&D level [11]. Integrated resource planning requirements at many utilities make this type of forecasting a necessary step, particularly in long-range studies. Recent advances in data collection have brought down the cost and time required to gather data for simulation, but they still require more investment in manpower and computation than trending [11].

Figures 2 and 3 show the basic concepts behind the simulation. Each small area in the grid is classified by the types and amount of "land-use" present in its area. Land-use corresponds to customer classes -- residential, commercial, and industrial.

Future changes in land-use -- projections of where, how much, and what type of new residential, commercial, and industrial customers will occur on a small area basis, are forecast based on one of several land-use econometric simulations [11]. Such simulation works well as a spatial forecast method because the different customer classes such as residential, commercial, industrial, and various subclasses within them, have identifiable, forecastable patterns in their locations. As examples, 93% of all industrial growth in large urban areas occurs within 1/4 mile of a railroad line, and residential load growth is 85% more likely to occur between 1/8 and 2 miles of a major shopping center, than either closer or farther than that distance [1, 4, 11]. Given that growth will occur, areas with these and other similar characteristics are identified by the simulation method and given a high probability of growth. This yields a small area map of where future customers will be located.

This land-use map is converted to electric load by using kW per acre load curves on a land-use class basis, or alternately, an end-use class by class model that works on a kW/acre basis [EPRI 2548-1]. An important aspect of the land use approach is that the different classes -- residential, commercial, industrial, and subclasses within each -- have fundamentally different load characteristics. Each has a different daily load curve shape, load density, power factor, diversity with peak, etc. While individual customers within each class vary in these factors, their characteristics are similar, and their differences explainable based on adjustment with factors that are usually obtainable in either the small area land-use or load data [1, 4, 11].

In end-use models, the hourly peak-day demand of each customer class is disaggregated by basic end-use categories such as heating and lighting. Changes in these individual classes are forecast, and these forecasted future end-use hourly load curves are then added to obtain the forecasted total customer peak-day load curve. In this way, the impact of changes in market penetration (e.g., an increase in water heating usage from 36% to 43% of residential customers), increases in efficiency (e.g., lighting), and changes in usage patterns (e.g., more homeowners using set-back thermostats) can be assessed simultaneously [11]. A complication, often difficult to model and verify as accurate, is the representation of coincidence of load at the T&D level, which causes each customer class's load factor to decrease at the lower voltage portions of the system [11]. DSM studies, when a necessary part of T&D planning, build upon these end-use models, representing impacts of conservation, load control, and time-of-use programs as changes in the corresponding end-use load curves. Such models have proven useful, practical, and satisfactorily accurate when applied to T&D + DSM studies [10,11].

Thus, in the "land-use approach" both aspects of load growth -- spatial and temporal -- are simulated by an analysis that is segmented into customer or land-use classes. These two forecasts are then combined to obtain a "map" of electric load density (Fig. 1). Forecast results vary depending on the details of the specific method being used, but in general land-use-based forecast methods produce only 1/3 to 1/7 as much error in long-range (5+-year) distribution planning studies as the very best trending methods, but require up to twelve times as much data and one thousand times as much computation [1, 5, 8, 11].

In order to be useful for distribution planning, the grid of small area customer estimates must be converted to estimates of electric load. Although a few methods [1,4] use a kW/acre conversion factor for each land use class, the most accurate way has proven to be an end-use load curve model [6,10,11] as described in the text above. This produces an estimate of future daily

peak-day load for a typical customer in each class. Given this and the map of customer types by small area (Figure 2), a map of future load estimate (Figure 1) can then be produced.

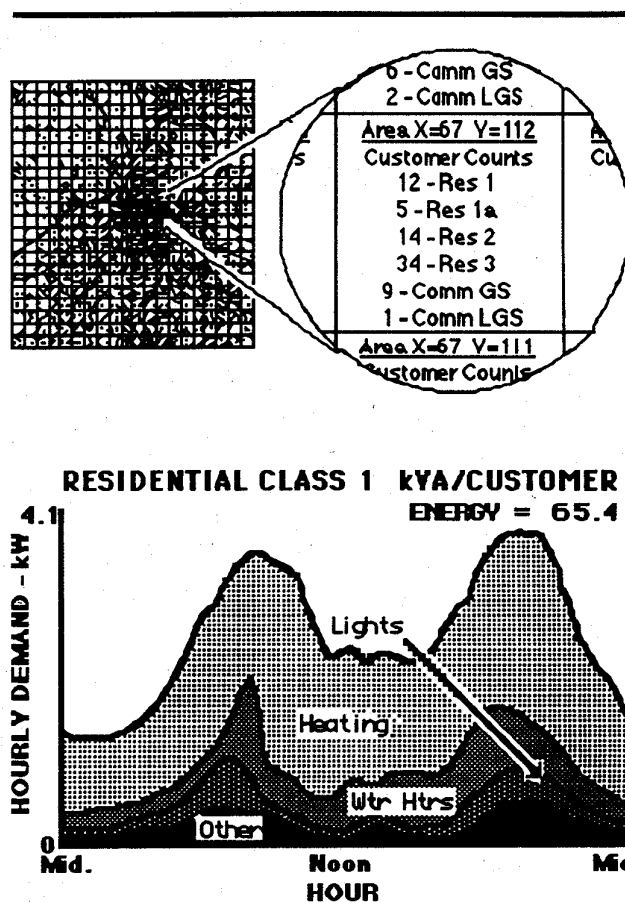


Figure 2. Simulation-based distribution forecasting methods work on a grid within which they first forecast future land-use inventory on a small area basis -- the number of residential, commercial, and industrial customers (usually by subclass within each category) in each small area as illustrated on the left. Methodologies are generally complex -- most apply some form of urban model in combination with pattern recognition [1, 4, 5, 8, 11]. In order to be useful for distribution planning, these small area customer count projections must be converted to estimates of electric load. A few methods [4,8,11] use a kW/acre conversion factor for each land-use class, the most accurate way has proven to be an end-use load curve model [4,10,11]. This produces an estimate of future daily peak day load for each class, on a per customer basis (right). Given this and the map of customer types by small area, a map of future load estimates (Figure 1) can be produced.

RURAL AREA FORECASTING

Despite its advantages in accuracy, the land-use-based simulation approach has a number of shortcomings when applied to forecasting in rural areas. These problems are not directly associated with the land-use concept itself, but rather with the grid small area structure.

The first problem is one of economy of use. A rural area is sparsely populated, so only a small fraction, usually less than two percent, of the

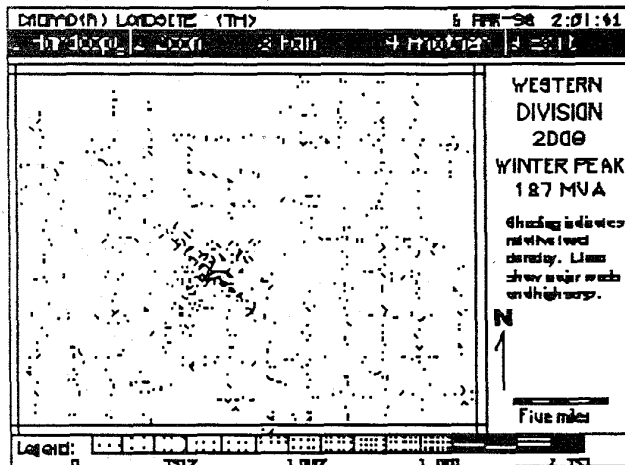


Figure 3. Small area map of rural electric load distribution. In an area that is shown in Figure 1, only about five percent as much load is distributed, a very low density. The vast majority of small areas (97%) have no load in them, in contrast to Figure 1 (11%).

small areas in the data grid have any load, or will ever develop any load, as shown in Figure 3. A grid-based method thus requires data storage, coding, and analysis for two to three orders of magnitude more data than is actually needed for the analysis. Alternatively, an involved "sparse matrix" approach that stores and keeps track of only the small areas which have loads can be used, but that makes it impossible to use various high-speed calculation methods that reduce computation when applied to a grid [11].

The authors believe that through various means the data and computer needs of grid-based programs could be modified to streamline them for rural applications reducing their computation resources to reasonable levels[4,11].

However, there is an additional problem with grid based simulation when applied to rural areas – one of accuracy. It is not nearly as accurate as when applied to urban areas. Figure 4 summarizes results from two test forecast cases, results which are representative of the vast majority of situations the authors have observed. While a grid-based land-use/small area grid method is more accurate than trending regardless of whether applied to urban or rural areas, it gives 30% more error in rural forecasts than in urban forecasts. Non-land-use based methods, though less accurate overall, display no difference in rural and urban settings. The authors

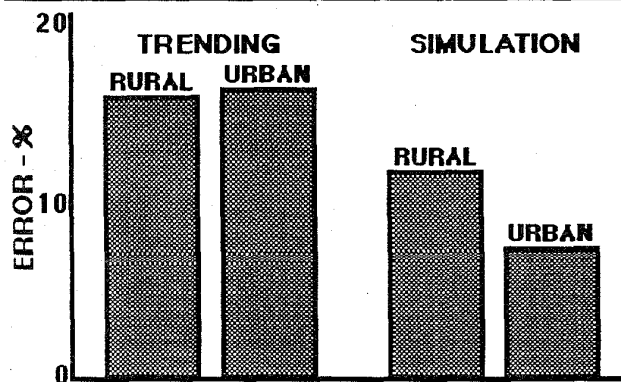


Figure 4. Forecast error for two distribution forecasting techniques, Methods 15 and 12 as described in Chapter 2 of *IEEE Power Distribution Planning Tutorial*. These are generally acknowledged as the most accurate trending and simulation methods in wide usage, respectively. Here, the methods were applied to two test problems, an urban system growing from 2250 MW to 4409 MW peak load over 20 years and a rural system with 204 MW peak load growing to 285 MW over 20 years (Test Utilities A and B in that Tutorial). Error measure is percent of distribution loads misforecast sufficiently to most likely cause planning errors [5].

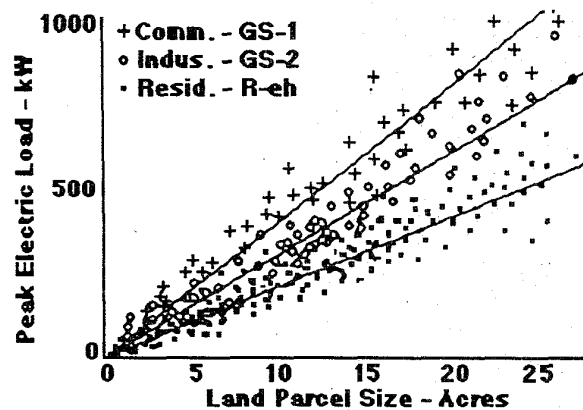


Figure 5. Scatter diagram of peak load versus land parcel size for 254 randomly selected properties/land parcels in Houston, Texas, (1980). A linear relationship between parcel size and load on a class-by-class basis is quite apparent. Land area, given *a priori* knowledge of the land-use type, is quite clearly a very good predictor of load level.

interpret these results to mean the following: that forecasting distribution loads in rural regions is not inherently more difficult than in urban areas (otherwise trending would show a difference in accuracy between the two), but that simulation, as applied in the past, has some flaw or inappropriateness with regard to rural settings.

The problem, at least in part, can be seen by examination of distribution load data as it appears in a small area grid. Shown in Figure 5 is a plot of land parcel size versus peak annual electric load for 254 individual non-uniformly sized parcels of land in a major city (Houston, 1980). Most of the points fall along three major lines which are residential, commercial, and industrial class load densities, respectively. Within each class, load correlates quite linearly with land area. For the most part, individual parcel variations from the linear relationship within each class can be explained by acknowledging various "surround" and "locality" factors described in detail elsewhere [2, 11]. The important aspect here is that load and land parcel size are relatively linear within an urban and suburban environment.

Shown in Figure 6 is a similar plot of 2,190 land parcels in a predominantly rural region. There are four classes – residential, commercial, industrial, and farm. Within each class, correlation is much poorer than in Figure 5,

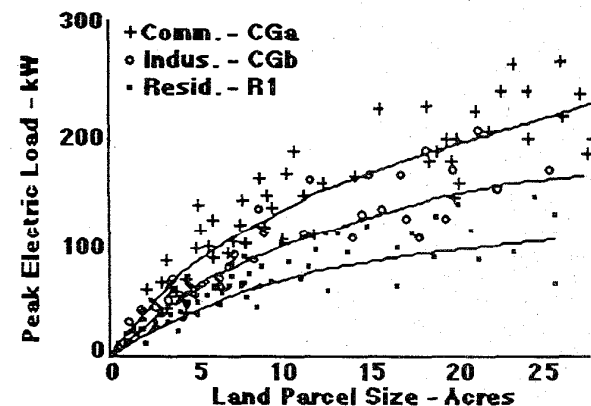


Figure 6. Scatter diagram of peak load versus land parcel size for 137 randomly selected properties/land parcels in rural areas outside of Houston, Texas (1980). The relationship between parcel size and load on a class-by-class basis is quite apparent is best represented as proportional to the square root of parcel size (plotted lines).

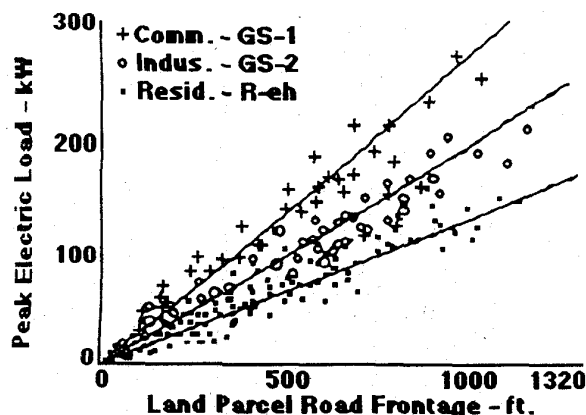


Figure 7. Representation of the data in Figure 6 as a function of road frontage of the land parcels, shows a good linear relationship. Apparently in rural areas (at least this one) demand level is more proportional to road frontage than parcel area

and the relationship that is evident between parcel size and load appears to be non-linear. The authors' study of this revealed that load corresponds more closely with the square root of land parcel size than land parcel size itself. This is shown in Figure 7, where Figure 6 data is replotted as load versus square root of land parcel size. The improved relationship, now apparently linear although having a wider range of deviation, is quite clear. Existing land-use based simulation methods applied on a grid basis implicitly relate electric load to land area in a linear manner. Given the differences shown in Figures 6 and 7, it is not surprising that these methods give better results when applied to urban areas than in rural applications.

Some Comments on the Inverse Quadratic Relationship and the Importance of a Linearity in Simulation Models

A weakness of small area forecasting methods, often not recognized, is that they assume linearity between area and load. Since this is in fact a characteristic of urban areas – a 100 acre residential development has twice the load of a 50 acre development, an 90 acre mall has thrice the load of a 30 acre retail development – this characteristic has never been viewed as a problem.

However, this implicit assumption means that such models cannot easily model non-linear area-to-load relationships of the type identified from the data in Figure 5. In a rural areas, the load in a residential area composed of 20 small areas is about 1.4 times that. Given the structure of simulation methodology, the authors judged that it would not be straightforward to modify simulation methods to model this relationship.

The authors do not mean to imply that small area simulation forecast methods cannot be modified in some manner so they could handle nonlinear relationships, but only that they could think of no simple, apparently accurate modification that did not entail creating what was essentially a completely new type of forecast methodology.

Instead, the authors believe that a better way to improve simulation was to find something other than area upon which to base the "small units."

Road Frontage

Figure 7 shows the rural land parcels from Figure 6 replotted in a slightly different form, as load versus road frontage. Road frontage is total length of all edges of a parcel that border on a paved public road. Linearity between this parameter and load is quite clear.

As demonstrated by the data in Figure 7, in rural areas road frontage is a better indicator of the potential for a parcel of land to develop electric demand. This relationship makes sense to the authors (it is quite visible in Figure 3) and is clearly the correct interpretation. In a rural region only the land along roads is accessible and built upon. One hundred acres with 1,000 feet of road frontage and ten acres with similar frontage have similar potential for load development.

The relationship between road frontage and electric load may strike the reader as obvious (the authors admit that they expected to find this relationship in the data), and it may seem that this examination was a lengthy exercise in proving a trivial point.

However, road frontage is also important in urban areas – land simply never develops unless there is access to it and road frontage is needed under any circumstances. The difference is that in urban areas almost all land is close enough to one road or another that sufficiency can be assumed. Frontage is of secondary importance and only the impact of certain major roads needs to be acknowledged [11]. The data shown here and similar studies conducted by the authors on other rural/urban comparisons provide quantitative verification of this concept and the fact that area-to-load linearity does not apply in rural areas but that road frontage does.

ROAD FRONTAGE-BASED SIMULATION LOAD FORECASTING METHOD

Based on the results presented above, the authors constructed a forecasting method oriented around "small road links," rather than small areas. This new method uses *exactly* the 3-3-3 algorithm widely used in urban forecasting situations and described in detail elsewhere [4, 8, 11]. This method is precisely method thirteen in reference 11 (or with addition of a DSM load curve model in its end-use model, method 16).

Here, the 3-3-3 algorithm was implemented in a road link rather than small area basis by changing the basis for its data framework. The same computer program (a commercial software product named LOADSITE) code was used, simply set up to operate on a set of small road links rather than small areas. Using modern database representation methods, this proved to be a very simple procedure. There were *no* other differences, and the basic premises of the land-use simulation remain the same:

- load correlates with a measurable geographic quantity
- this correlation occurs within the customer (land-use) classes

This rural forecasting simulation method, which the authors refer to as RLLF (road-link load forecast) is based on a set of small-length road links, each link linked by data pointers to other links nearby. The links are all of uniform length – 1/8 mile to a side, so a 20-mile road segment is represented by 160 links. Links essentially have an implicit area, a length (road frontage) but no identifiable depth, since area actually doesn't matter.

A data structure using non-uniform facets (perhaps one 20-mile segment for a 20-mile road) would be more compact and lead to faster analysis. However, the authors used uniformly sized road links for two reasons:

- 1) Resolution. Load growth needs to be located with better accuracy than to within 20 miles. Use of 1/8 mile links locates them precisely enough. (See [8] for a discussion of small area resolution and forecast error, all of which applies in rural applications, too.)
- 2) Statistical Uniformity. All simulation methods, particularly those employing pattern recognition (the middle digit in the standardized nomenclature of algorithm classification is a "3", as in the 2-3-2 CEA model [4, 11]), have proven both faster and more accurate when some quantity about each area is absolutely constant. In the past, area has proven convenient and accurate as this non-varying quantity [1, 4, 11]. Uniform link lengths are analogous to this.

Table I compares important aspects of the RLLF with the grid based simulation (3-3-3) method from which it was derived. Implementation was straightforward, but several details deserve comment. First, just as all grid cells in most urban applications are the same size, all links in RLLF's database have the same length – 1/8 mile. Road segments longer than 1/8 mile are broken into multiple links, just as in grid models, shopping malls and other land parcels larger than the small area size are split into multiple cells. Road segments shorter than 1/8 mile are modeled as unuseable – i.e., unavailable for analysis. This is similar to the restricted land-use area classification used in grid-based methods.

Table I
Comparison of the RLLF Method
and the Grid Based Simulation Method

Grid based

LOCATIONAL DATA: Small area database consists of land-use (customer class) inventories for each small area, or grid cell, in the utility region.

LOCATIONAL REFERENCE SYSTEM: Cells are located with respect to other cells implicitly, based on their locations within the grid.

LOAD DATA: Program uses daily electric load curves on a class-by-class basis using end-use analysis of major load and appliance categories.

OVERALL APPROACH: Program first forecasts future customer class locations and densities for each cell, then forecasts each customer class's load curve shape, and finally uses those curves to convert the forecasted cell-by-cell land-use inventories to electric load.

CUSTOMER LOCATION: Forecast is done by pattern recognition that compares each cell's characteristics to the optimal pattern for each land-use class's development, e.g., commercial develops near major road intersections, industrial develops near railroads.

DISTANCE: Measures needed by the pattern recognition are calculated as straightline distance between points in the database.

ELECTRIC LOAD: Is handled by projecting future per daily load curve shape for each customer class on a per acre basis.

FINAL FORECAST: Is produced by combining for each cell the projected land use, converted by class to load on a per customer basis.

RLLF

LOCATIONAL DATA: Small area database consists of land-use (customer class) inventories for each "link" in the road network.

LOCATIONAL REFERENCE SYSTEM: links are located with respect to other links explicitly, using data pointers to neighbors.

LOAD DATA: Exactly the same as the grid approach, on a class-by-class basis using end-use analysis of major load and appliance categories.

OVERALL APPROACH: Program first forecasts future customer class locations and densities for each link, then forecasts each customer class's load curve shape, and finally combines those customers on a link-by-link basis with class load curves to produce the small area loads.

CUSTOMER LOCATION: Forecast is done by pattern recognition that compares each link's characteristics to the optimal pattern for each land-use class's development, e.g., commercial develops near major road intersections, industrial develops near railroads.

DISTANCE: Measures needed by the pattern recognition are calculated by a tree-walk procedure following links through the database.

ELECTRIC LOAD: Is handled by projecting future per daily load curve shape for each customer class on a per mile basis.

FINAL FORECAST: Is produced by combining for each link the projected land use, converted by class to load on a per customer basis.

COMPARISON TESTS WITH OTHER DISTRIBUTION FORECASTING METHODS

The RLLF technique was compared to two other load forecast methods in a series of forecasting tests on both rural and urban areas. The two other forecast methods were LTCCT trending, and a grid-based land-use (3-3-3) simulation method, methods 15 and 12, as described in reference 11. These two methods are respectively the most accurate trending method and the most accurate simulation (most accurate overall) among sixteen methods examined in that Tutorial.

The three test problems included two rural regions, while the third was a medium-size (pop. 125,000) urban area. In each case the test involved forecasting known 1977, 1979, and 1984, and 1989 summer peak small area loads, working from 1974 and earlier data – forecast periods of 3, 5, 10, and 15 years ahead, which were then compared to actual loads in this years. Tables II and III summarize the test results of the three algorithms, Figure 9 plots average forecast accuracy for the three problems.

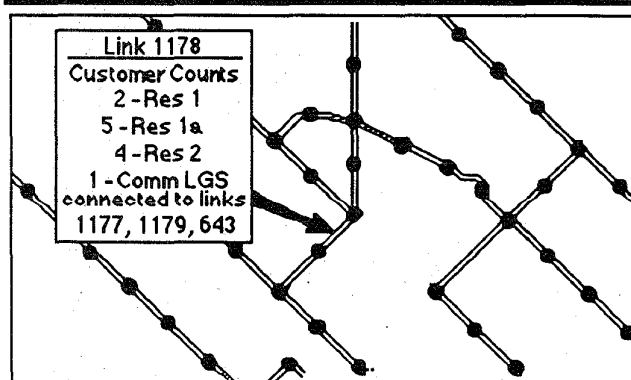


Figure 8. Road link representation in RLLF method consisted of "small link" data sets each giving the customer class usages along that road link and also having pointers to other links in the data set. Over half of the data stored in the tests consisted of data pointers to other links, a major computational overhead (by contrast grid-based methods implicitly tie neighboring small areas together by grid location). Regardless the RLLF method was still more efficient in rural applications by a factor of ten.

Accuracy of all three methods was determined by subtracting the forecast 1977, 1980, 1985, and 1990 small area loads from the known actual values for those years. Forecast error is evaluated by the distribution system impact error measure, U, an estimate of the percentage of distribution capacity expected to be mis-planned because of the forecast errors, as estimated by the Spatial Frequency Analysis method of error evaluation [11]. As an error measure for *distribution planning applications* this has numerous advantages over RMS and similar statistics AAV [5].

In the two rural test problems, both simulation methods gave much less error than the trending method. However, the RLLF method delivered better accuracy than the grid-based simulation, with error in ten-year ahead forecasts fully 40% lower, and in the range of the best error levels achieved.

Table II
Forecast Test Accuracy of
Three Forecasting Methods
in Rural Applications

Error Measure U - %

Test Problem 1		3 years	5 years	10 years	15 years
Method	Base				
RLLF	1	3	6	9	16
Trending	0	5	8	19	32
Grid-Sim	4	4	7	13	22
Test Problem 2		3 years	5 years	10 years	15 years
Method	Base				
RLLF	0	3	7	10	16
Trending	0	3	7	18	37
Grid-Sim	4	6	8	12	21
Test Problem 3		3 years	5 years	10 years	15 years
Method	Base				
RLLF	6	8	11	14	19
Trending	0	4	7	19	35
Grid-Sim	1	2	3	8	12

Table III
Summary of
Three Forecasting Methods

Test Problem 1 - Rural, 5400 square miles, 34,000 customers, 217 MVA

Method	# Areas ¹	Data ²	CPU ³	Man-hrs ⁴	Error ⁵
RFLF	4,100	24 Mb	9.9	134	9%
Trending	180	280 kb	47	16	19%
Grid-Sim	86,400	280 Mb	29.3	200	13%

Test Problem 2 - Rural, 17,000 square miles, 23,000 customers, 190 MVA

Method	# Areas ¹	Data ²	CPU ³	Man-hrs ⁴	Error ⁵
RFLF	3,800	18 Mb	8.3	110	10%
Trending	250	280 kb	.65	12	18%
Grid-Sim	69,200	273 Mb	67.3	200	12%

Test Problem 3 - Urban, 860 square miles, 127,000 customers, 823 MVA

Method	# Areas ¹	Data ²	CPU ³	Man-hrs ⁴	Error ⁵
RFLF	51,600	334 Mb	77.3	420	14%
Trending	800	180 kb	3.5	18	19%
Grid-Sim	21,500	63 Mb	21.3	200	8%

¹ Number of small areas, road links, or trending feeder portions. ² Disk storage space. ³ CPU time in minutes on a 386-33Mhz PC, programs in "C". ⁴ Estimated data preparation and analysis man-hours for an experienced user. ⁵ Error measure U, percent error, ten year forecast (1974-1984).

in urban forecasting situations. In the urban case, RLLF's accuracy exceeds that of the trending method by a considerable margin (30%) but is still 35% more than that of the grid-based method. Basically, the difference in land area versus road link relation to load is turned around in this problem, and the road link method is at a disadvantage.

Resource requirements for the RLLF method in rural forecasting applications are considerably less than those required by grid-simulation method. Computer storage and data requirements are more than an order of magnitude less. Computer processing times fall by a factor of from three to seven. Data gathering and other manpower requirements are cut nearly in half. Both simulation methods require considerably more resources than those required by trending. However, the RLLF simulation has what the authors believe are reasonable and affordable levels of data, computer, and manpower requirements (equivalent to what simulation demands in urban applications).

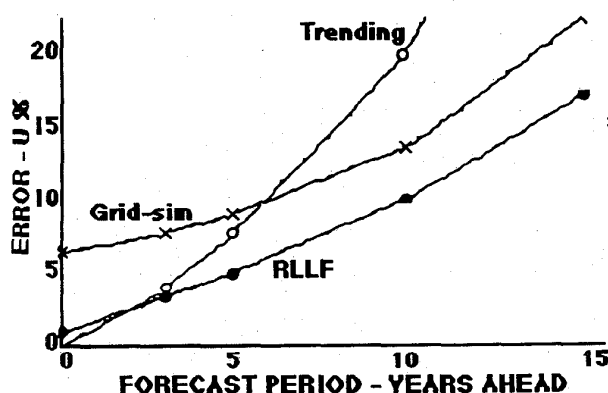


Figure 9. Average error for the two rural test cases, given by each of the three methods. Error in year zero is the "calibration" or "fitting" error of the simulation when adjusted as well as possible to metered feeder and substation loads [4, 11].

CONCLUSION

This paper has investigated distribution load forecasting, concentrating on why grid-based simulation methods have failed to achieve accuracies in rural areas comparable to those they achieve in urban areas. Examination of rural electric load revealed that if broken into customer classes, load density corresponds more nearly to the square root of land parcel size, rather than land area, as in urban regions. This is apparently the cause of grid-based simulation inaccuracies when applied to rural areas.

Further investigation showed that in rural areas, load corresponds very linearly to road frontage rather than land parcel size. A new forecast method utilizing a standard distribution load forecast simulation algorithm but oriented around the concept of "small road-links" rather than small areas, proved to be very easy to develop, a straightforward database modification of an existing program.

This new method uses considerably lower data, computer, and manpower resources than grid-based methods in rural applications, and more importantly, delivers error levels thirty to forty percent less than grid-based methods do in these cases - accuracy levels comparable to the best achieved by simulation methods in urban applications.

Equally important in the authors' opinion, this new framework of application can be implemented without major changes to existing numerical method, allowing not just the authors' algorithm, but very likely most other simulation methods to be modified easily for rural application.

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THE AUTHORS

H. Lee Willis, (F) is Senior Principal Consultant for ABB Systems Control Company, as well as Director of Technology for ABB's Automated Distribution Division in Raleigh, NC. He is currently secretary of the IEEE Power System Engineering Committee. He has authored or co-authored over 200 technical papers, as well as the books *Introduction to Integrated Resource T&D Planning*, and *Planning Power Distribution Systems*.

Linda A. Finley, (SM) is Director of Technical Services at Snohomish PUD, Everett, WA. She is chairman of the IEEE Demand Side Management Subcommittee.

Michael J. Buri, (M) is a Project Supervisor with ABB's Automated Distribution Division in Raleigh, NC.